

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		Both correct for 1 mark X= carbon dioxide / CO ₂ Y = oxygen /O ₂ / accept O	1			1		
		(ii)		Capillary	1			1		
	(b)	(i)		6 000 (1) 1 000 (1)		2		2	2	
		(ii)		Deeper breaths/more heavily / increased depth / larger volume / bigger breaths (1) Faster/quicker/increase <u>rate</u> / more frequent (1) Allow increases with no qualification for (1)		2		2		
		(iii)		Volume of oxygen = 21-16 = 5% (1) × 40 000 (1) = 2 000 cm ³ (Yes she has) (1) OR 21% × 40 000 = 8 400 (1) 16% × 40 000 = 6 400 (1) Difference is 2 000 (so yes she has) (1)			3	3	3	
	(c)	(i)		without air/oxygen	1			1		
		(ii)		Advantage – quick/fast (release of energy) / allows muscles to carry on (working when oxygen levels are low) (1) Disadvantage - produces small quantities of energy / produces lactic acid / causes cramp / oxygen debt (1)	2			2		
				Question 6 total	5	4	3	13	5	